Commuting interval factors and the crossing anchor

Human Compression project: bounded L044 proof component

Use the complete block-deficit setting: $G \le A_n$ is a transitive string C-group, $n \ge 12$, every maximal omission is intransitive, and a unique eligible i-edge joins $a \in A$ to $b \in B$, the two G_i-orbits. Their two-block systems have block sizes k, l, respectively. Write $L = \langle \rho_j : j < i \rangle$ and $R = \langle \rho_j : j > i \rangle$. These commute, and $G_i = LR$.

Lemma 1 (An anchor fixes the whole commuting orbit)**.** *If every generator of R fixes a, then R fixes A pointwise. The symmetric assertion holds for L and b.*

Proof. For $l \in L$ and $r \in R$, commutation gives $(a^l)^r = (a^r)^l = a^l$. Moreover $a^{lr} = a^l$, so the L-orbit of a is its whole G_i-orbit A. Every element of R therefore fixes every point of A. □

Lemma 2 (No regular orbit away from its anchor)**.** *Suppose $\alpha = 0$, $\beta = 1$, $\gamma = 0$. The unique regular four-point orbit D of $M \cong C_2^2$ must contain the crossing endpoint in its original G_i-orbit.*

Proof. Suppose instead $D \subset A$ and $a \notin D$. Both anchors then lie in paired M-orbits. Only the literal neighbors ρ_{i-1}, ρ_{i+1} can exchange blocks: every other generator fixes both anchors. The two independent block-image generators are precisely those neighbors. If they both exchange the blocks of either orbit, they act identically on its anchor pair. Every other generator fixes its opposite anchor, using commutation with one neighbor or, for either adjacent exception, with the opposite neighbor. The pair would be G_i-invariant, impossible. Thus the actual block-image basis is pure. After reversal, $\tau_A = \rho_{i-1}$ exchanges only the blocks of A, and $\tau_B = \rho_{i+1}$ exchanges only the blocks of B.

On each paired M-orbit in A, projection to its two blocks is an equivariant bijection. The element τ_B consequently fixes that pair pointwise, in particular fixing a. All other generators of R fix a by the unique-crossing property. Lemma 1 therefore makes R fix A pointwise. This is impossible: on the regular M-orbit D, the nonidentity element τ_B has two transpositions. □

Lemma 3 (Exact support separation for paired rows)**.** *If $\alpha = \beta = 0$, the literal pure block-image basis established in the two-swap saturation note gives L supported on A and R supported on B. Their actions there are faithful transitive string C-markings with their original generator orders.*

Proof. The right neighbor fixes A pointwise, and all other generators of R fix a. Lemma 1 makes all of R trivial on A. The same argument makes L trivial on B. Faithfulness follows from these being the only two original orbits. Transitivity follows from $G_i = LR$; string relations, independence and IP are inherited by the unchanged intervals. □

Lemma 4 (The paired-row rank bound)**.** *If $\alpha = \beta = 0$, then $r \le k + l - 1 = (n-2)/2$.*

Proof. The marked pure-basis and parity arguments in the two-swap note give even $k, l \ge 4$. By Lemma 3, the left interval is supported on A and transitive there; the right interval is supported on B and transitive there. Let their ranks be d_A, d_B, so $r = d_A + d_B + 1$.

The left interval ends with $x = \rho_{i-2}$ and $\tau_A = \rho_{i-1}$, both present. All its earlier generators commute with τ_A and fix a, so act faithfully and diagonally on the other $k-1$ row pairs. Their independent length is at most $k-2$ by Whiston. Hence $d_A \le k$. Suppose equality holds. The earlier family then generates the full S_{k-1}, with a maximal independent involutory string marking.

For $k \geq 6$, the accepted L047 classification makes this the ordinary Coxeter path: $k-1$ is odd, so the outer degree-six exception is absent. Its near-end omission is the natural S_{k-2} fixing the anchor row and one additional row v. The generator x centralizes that omission, preserves the two original blocks and fixes a. It is consequently the identity on the block containing a, and on the other block it is the transposition exchanging a' with v', since it must move a' to another row. Its support is contained in A by Lemma 3. It is therefore odd, impossible.

For $k = 4$, denote the far and near tail generators by p, q. Choose row labels $0, 1, 2, 3$ and $0', 1', 2', 3'$ with $a = 0$, so

$$p = (1\ 2)(1'\ 2'), \quad q = (2\ 3)(2'\ 3'), \quad \tau_A = (0\ 0')(1\ 1')(2\ 2')(3\ 3').$$

Since x centralizes p, fixes 0, moves $0'$ to another row, and is even, its only possibilities are

$$x = (1\ 2)(0'\ 3') \quad \text{or} \quad (1'\ 2')(0'\ 3').$$

In either case $\langle q, x, \tau_A \rangle$ is transitive on all eight points. This follows directly from its permutation graph: contract the four τ_A pairs to row vertices. The q edges join rows $2, 3$, and the x edges join $0, 3$ and $1, 2$. These three row edges form a spanning path, and each contracted pair is itself joined by τ_A. The original graph is thus connected, proving transitivity without an order computation.

Let j be the index of p. The global omission G_j contains this transitive subgroup on A, the unchanged entire right interval transitive on B, and ρ_i joining a to b. It is therefore transitive on $A \cup B$, contradicting the standing hypothesis on every maximal omission. This excludes $d_A = k$ also at $k = 4$.

We have proved $d_A \leq k-1$. The symmetric proof gives $d_B \leq l-1$, whence $r \leq k+l-1$ as asserted. □

Scope and attribution. The source problem and block-image framework are from Cameron, Fernandes, Leemans and Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 5.1. The elementary commuting-factor argument supplies a short new proof in that framework. The complete accepted block-deficit and new frozen two-swap components remain paid dependencies. Together with the anchored regular-orbit proof, Lemma 2 closes $\beta = 1$ for two blocks on each original orbit. Lemma 4 closes the remaining $\alpha = \beta = 0$ paired-row case. Combined with the earlier $\alpha = 1$ endpoint results and the exact deficit formula, these prove the rank bound when both original orbits have two blocks. Mixed block-count and other source branches remain. Full-source recovery, new computation, new formalization and new-question credit are not claimed.